Table 1: SIR DDABM components, assumptions, and data sources.

ABM Component	Sub-category	Element	Value / Assumption	Data Source
Agents	State	Disease states	S, I, R	—
	Attributes	Demographics	Age, gender, occupation	Census datasets (static)
	Population	N (total agents)	Fixed 10,000; census-derived	Census datasets (static)
	Behaviors	Movement patterns	Mobility-informed; location-dependent	GPS data (dynamic)
Interaction Rules	Transmission	S → I	Contact with infected; prob. = β	Public health reports (dynamic)
	Recovery	I → R	Stochastic; prob. = γ; permanent immunity	Public health reports (dynamic)
	Agent–Agent Interaction	Contact patterns	Mobility-informed; location-dependent	GPS data (dynamic)
Environment	Locations	Activity contexts	Home, work, school	GPS mobility data (dynamic)
Other Key Parameters				
Interventions	NPIs	Measures	Social distancing, masking, vaccination; applied as β multipliers	Health dept. guidelines & surveys (dynamic)
β, γ	Transmission & recovery rates	Parameters	Estimated via calibration	Public health reports (dynamic)
Time step	Temporal resolution	Step size	1 day	—
Duration / Stop	Simulation length	End condition	100 days; stop when I = 0 or max duration reached	—

Table 2. Data-driven workflow summary for SIR DDABM.

Stage/	**Details**	**Data / Input**	**Method**	**Output / Role**
Data Inputs & Pipeline	Collection	Census, GPS, health reports	Static extraction; near real-time API feeds	Raw datasets for initialization & updates
	Pre-processing	All collected datasets	Consistency checks; multiple imputation (missing counts)	Cleaned data for ABM initialization
	Analysis	Pre-processed data	Statistical estimation; contact-rate derivation	Initial $\hat{\beta}$, $\hat{\gamma}$ estimates; contact patterns
Model Evaluation	Calibration	Historical infection time-series	Bayesian inference over (β, γ)	Calibrated β^*, γ^*; uncertainty bounds
	Output validation	Held-out surveillance data	RMSE/MAE vs. observed S/I/R counts	Quantified accuracy on unseen data
	Scenario validation	Post-intervention surveillance	Simulated vs. recorded post-NPI trends	Confirmed model responsiveness to NPIs
Scenarios (Simulation Experiments)	Baseline	Calibrated β^*, γ^*; no NPI	Standard SIR ABM run	Reference epidemic curve
	Social distancing	Reduced contact rate	$\beta \times (1 - \text{efficacy})$	Flattened curve; delayed peak
	Vaccination	Vaccination rate (surveys)	S → R at rollout rate	Reduced susceptible pool
	Combined NPIs	Distancing + masking + vaccination	Multiplicative β reduction; combined adherence probs.	Key scenario for DT-informed policy recommendation